



DELAGUA IMPROVED COOKSTOVE GROUPED PROJECT



Project title	DelAgua Improved Cookstove Grouped Project
Project ID	3699
Monitoring period	01 January 2023 – 31 December 2024
Original date of issue	14-February-2025
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VCS Standard Version	4.7
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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The grouped project involves dissemination of energy-efficient, improved cookstoves (ICS) to replace existing traditional cookstoves in domestic households of Rwanda. The grouped project has only one project activity instance (PAI-1) till the end of this monitoring period.

The distribution of the project ICSs begun on 04-November-2022 under the PAI-1. This date is also the start date of the grouped project and the start date of crediting period for this project. Till the end of current monitoring period, total 334,944 ICSs have been distributed throughout Rwanda. During the current monitoring period, the project has achieved GHG emission reduction of 1,911,382 tCO_{2e}.

Technology/ Measures

The ICSs disseminated under the grouped project are based on a “rocket stove” and designed to improve fuel combustion and heat transfer. The project stoves improve heat transfer to cooking pots through the insulated walls and design of the stove top when compared to traditional cooking methods over open- and three-stone fires. The ICS results in the reduction of fuel consumption and improvement in levels of Indoor air pollution for project users (reducing smoke, black soot, and particulate matter (PM) emissions). The reduction in consumption of cooking fuel also reduces equivalent greenhouse gas (GHG) emissions attributed to the use of non-renewable biomass.

For the current monitoring period, the PAI-1 includes dissemination of Burn Kuniokoa G2 stoves, a tier 3 cooking devices with the following specification:

Name of the ICS	Kuniokoa Generation 2		
Manufacturer	Burn		
Type	Natural Draft		
Material	Stainless Steel		
Dimensions	32.1 cm Height, 28.2 cm Diameter		
Thermal efficiency	With Pot skirt – 51.3%, Without pot skirt – 43.4%		
Power Output	2.9 kW		
Lifetime	10 Years		
Photo			

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
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Validation	04-November-2022 - 03-November-2029	VCS	Earthood Services Private Limited	1 st Crediting Period (7 Years)
Verification	04-November-2022 - 31-December-2022	VCS	Earthood Services Private Limited	0.158 year (58 days)
Verification	1-January-2023 - 31-December-2024	VCS	SustainCERT SA	Two Years

1.3 Sectoral Scope and Project Type

Sectoral scope¹	03 – Energy demand
Project activity type	Type II – Energy efficiency improvement projects

1.4 Project Proponent

Organization name	DelAgua Trans Africa Stoves Pte. Ltd. (DATAS)
Contact person	Neil McDougall
Title	Chairman, Director
Address	38 Beach Road, South Beach Tower, #29-11 Singapore 189767
Telephone	+352 661 939 304
Email	vcsprojects@delagua.org

1.5 Other Entities Involved in the Project

Organization name	DelAgua Health Rwanda (Implementation) Limited
Contact person	Rory McDougall
Role in the Project	Distribution, operational and monitoring services
Title	Commercial and Finance Director
Address	#59 Kg 9 Ave, Gasabo, Kigali, Rwanda
Telephone	+250 788 386 128
Email	vcsprojects@delagua.org

1.6 Project Start Date

Project start date	04-November-2022
Justification	Date of the first distribution of ICS

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	VM0050 - Energy efficiency and fuel-switch measures in cookstoves	1.0
Tool	CDM Tool:30 ²	TOOL30: Calculation of the fraction of non-renewable biomass	4.0
Guideline	-	Guideline: Sampling and surveys for CDM project activities and programmes of activities	4.0
Standard	-	Standard: Sampling and surveys for CDM project activities and programmes of activities	9.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.12 Sustainable Development Contributions

The grouped project activity involves dissemination of energy-efficient, improved cookstoves (ICS) to replace existing traditional cookstoves in domestic households in Rwanda. The GP is contributing towards sustainability, which is also verifiable through the SDG indicators selected for the specific activities that are going to be monitored for their fulfilment.

The grouped project has only one project activity instance (PAI-1) till the end of this monitoring period. A summary of sustainable development contributions for the PAI – 1 is provided in the following table:

SDG Targeted	SDG Indicator	Sustainable Development Benefits	Nationally stated Sustainable
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			Development Priority ³
SDG 1	1.4.1 Proportion of population living in households with access to basic services	Cooking is widely recognized as an essential service, given that food is a fundamental necessity for humans and other living beings. Therefore, improved access to cooking facilities is likewise deemed a basic need and essential. The project by distributing ICS provides access to basic services.	Yes
SDG 3	3.9.1 Mortality rate attributed to household and ambient air pollution	Evidence: ICS distribution Database monitoring survey results The indoor pollutants emissions attributed to inefficient burning of solid biomass fuels in the beneficiary households are reduced. The project ICS aids improved combustion of solid biomass fuel thereby reducing smoke and Particulate matter (PM) emissions.	Yes
SDG 4	4.7.1 Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment	Evidence: ICS distribution Database monitoring survey results The project activity educates and trains the team involved in the distribution of ICS and educates the end user HHs about sustainable cooking practices. With this, it can be said that the local population is learning sustainable practices and a healthy lifestyle without spending anything.	Yes
SDG 5	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Evidence: 1) Distribution Database 2) CHW training Data The project ICS saves time in cooking activities (by faster cooking) as well as save time in fuel procurement (reduced collection time) due to reduced fuel consumption. The proportion of HHs confirming the reduction in amount of time spent on collection of firewood and cooking time during the survey, multiplies with the total number of households to get the total number of impacted HHs.	Yes

³<https://www.minecofin.gov.rw/index.php?eID=dumpFile&t=f&f=15326&token=3f50e67804168295fc49c40ef03de1718f5bdaf2>

SDG 7	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Evidence: Monitoring Survey for VCS GP includes questions on time spent on collection of fuel. The project activity by distributing ICS provides access to clean cooking with primary reliance on clean fuels and technology.	Yes
SDG 8	8.3.1 Proportion of informal employment in total employment, by sector and sex	Evidence: ICS distribution Database monitoring survey results As part of the project activities (distribution and education campaign) DelAgua employs local community members.	Yes
SDG 13	13.0 Total greenhouse gas emissions per year	Evidence: The salary slips for the CHWs act as the data source since these are not on DelAgua's payrolls. However, there are different data sources (number of mobile phones assigned to individuals during distribution, number of CHWs trained for a specific distribution event), which are used to total the CHWs working for DelAgua. The project by replacing the inefficient traditional stove with the energy-efficient ICS shall reduce equivalent GHG emissions for meeting similar thermal energy needs as that in the baseline.	Yes
SDG 15	15.4.2 Mountain Green Cover Index	Evidence: Monitoring report and ER Calculation Sheet The implemented project activity leads to a decrease in deforestation in the mountainous forest ecosystem. This eventually leads to an increase in the mountain green cover index compared with the baseline. Evidence: Registered PD, MR and ER Sheet	Yes

Table 1: Sustainable Development Contributions

Row	SDG targ	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	1.4	1.4.1 Proportion of population living in households with access to basic services	Implemented activities to increase	Cooking is widely recognized as an essential service, given that food is a fundamental necessity for humans and other living beings. Therefore, improved access to cooking facilities is likewise deemed a basic need and essential. For PAI-1, the total number of ICS distributed is 334,944 i.e., 334,944 HHs receiving the project ICS now have access to basic cooking services.	Till the end of this monitoring period, 334,944 households now have access to basic cooking services when compared with the baseline.
2)	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	Mortality rate attributed to HH and ambient pollution will decrease for those HHs using the DelAgua improved cookstove. The improved design specifications of the project ICS model(s) ensure there is much less pollution in the households as compared to the baseline. The survey results demonstrate that 99.21% of households confirmed reductions in the smoke level. So, a total of 332,286 households now has less risk towards respiratory ailments due to smoke inside their homes due to the implementation of this project	Till the end of this monitoring period, 332,286 households now have less risk towards respiratory ailments due to smoke inside their homes due to the implementation of this project as compared to the baseline.

3)	4.7	4.7.1: Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment	Implemented activities to increase	The project activity educates and trains the team involved in the distribution of ICS and educates the end user HHs about sustainable cooking practices. With this, it can be said that the local population is learning sustainable practices and a healthy lifestyle without spending anything. i) The number of CHWs that were educated (trained for distribution, 6-monthly household visits, surveys, and training on sustainable cooking practices to be disseminated to end user households) by the Grouped Project (for PAI-1) by the end of M.P.-2 are 3,239. ii) The number of end users per household that are educated for stove usage and apply sustainable cooking practices are equal to the exact number of households where project ICS are distributed. PAI-1: 334,944 people educated.	Till the end of this monitoring period, 334,944 households were educated about stove usage and to apply sustainable cooking practices.
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4)	5.4	5.4.1: Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to decrease	Survey results show 96.83% of respondents stating that they have reduced the amount of time tending fire and collecting wood. Hence, 324,311 Households have reduced their time spent in fuel collection and cooking activities.	Till the end of this monitoring period, a total of 324,311 households were positively affected.
5)	7.1	7.1.2: Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase	The number of HHs with access to clean cooking has been increased with the implementation of this grouped project. A total of 334,944 HHs have received the ICS under the PAI-1. During the surveys, 99.08% of HHs confirmed usage of ICS. Thus, till the end of MP2, 331,873 distributed ICS are operational (i.e., These HHs have access to clean cooking when compared with the baseline).	Till the end of this monitoring period, 334,944 households now have access to clean cooking when compared with the baseline.
6)	8.3	8.3.1 Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase	As part of the project activities (distribution and education campaign) DelAgua employs local community members. To manage activities up to this point have created 3,239 informal employments.	Till the end of MP2, a total of 3,239 informal employment has been created with the project activity.
7)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	1,911,382 tCO ₂ e have been reduced during the current monitoring period.	Till the end of this monitoring period, a total of 1,970,855 tCO ₂ e emissions have been reduced by the project activity considering both MP1 and MP2.

8)	15	15.4.2: Mountain Green Cover Index	Implemented activities to increase	The implemented project activity leads to a decrease in deforestation in the mountainous forest ecosystem. This eventually leads to an increase in the mountain green cover index compared with the baseline. The amount of total firewood saved by the GP is 1,554,060 tons in MP2.	Till the end of this monitoring period, a total of 1,980,628 tons of firewood have been saved by the project activity considering both MP1 and MP2.
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1.13 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

The Stakeholder makeup has not changed since validation. Prior to the start of validation, PP conducted a comprehensive stakeholder consultation event as described in section 2.2 of registered VCS PD.

Stakeholder Identification	The Stakeholder makeup has not changed since validation. Prior to start of validation, hence this is not deemed applicable.
Legal or customary tenure/access rights	As the project is not a land use project, there are no legal or customary tenure issues or access rights that are impacted by the project.
Stakeholder diversity and changes over time	Refer section 2.2 of registered PD. No expected change has happened to the stakeholders' makeup
Expected changes in well-being	Household air pollution (HAP) is estimated to lead to 3.2 million ⁴ premature deaths globally each year, with the majority of those occurring in sub-Saharan Africa and Asia ⁵ . By providing improved

⁴ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

⁵ <https://ourworldindata.org/grapher/indoor-pollution-deaths-1990-2017>

	cookstoves and education on the benefits of clean cooking, the project will see positive changes in health. In addition, clean cooking has been shown to contribute towards various SDGs, such as gender equality and zero hunger, meaning the project will produce positive changes in well-being beyond health.
Location of stakeholders	Primary stakeholders are located in rural communities across Rwanda. District government officials are located in rural areas across Rwanda, whilst the Ministry of Environment is headquartered in Kigali (capital city.)
Location of resources	As this is not a land use project, this is not applicable.

2.1.2 Stakeholder Consultation and Ongoing Communication

Prior to the start of validation, PP conducted a comprehensive stakeholder consultation event as described in section 2.2 of registered VCS PD. The PP has already established a grievance mechanism / ongoing communication for stakeholders to raise any concerns about potential negative impacts of the project during project implementation.

Ongoing consultation	The PP has established a comprehensive grievance redressal mechanism to ensure that stakeholders have access to a transparent and easily accessible system for ongoing consultation and resolution of project-related concerns as detailed in section 2.1.4 below. The stakeholders are informed about the grievance register which is maintained at the local office locations of the PP i.e., in Kigali. Local people employed as field staff, and resource persons also serve as medium to escalate grievances received from the project beneficiaries in their respective zones to PP. Further, DelAgua has a toll-free helpline for the relevant stakeholders to connect with them in case they require any assistance. Any relevant concern received during operation of project activity is addressed according to its merit. The grievance register recording minor grievances raised during this monitoring period was shared with the VVB. Additionally, during the current monitoring period, the field team conducted household visits across multiple districts in Rwanda between 22 June 2023 and 20 December 2024. These visits were carried out as part of the additional ongoing engagement and communication process with end users. A total of 1,207 Community Health Workers (CHWs) participated in these visits, reaching 118,667 households with MQ barcodes. The visits served multiple purposes:
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	- Reinforcing key usage and maintenance practices through educational materials (flipbooks and posters). - Providing households with an opportunity to share feedback or raise any concerns. - Ensuring that households have access to grievance mechanisms, including direct communication with local field staff and information about the toll-free helpline. Any concerns raised were noted and addressed appropriately. These continuous visits support sustained use of the improved cookstoves and help maintain strong community relationships.
Date(s) of stakeholder consultation	The stakeholder consultation was conducted prior to validation at national level on 01-March-2023. Refer section 2.2 of the registered PD for details of consultation conducted prior to validation. Alongside this national level LSC, PP conducts local stakeholder meetings with before distribution of stoves in each sector/district or sign a collaboration agreement with the districts. Attendees in such meetings include local government official, alongside traditional local leaders from communities. This ensures the project is clearly communicated at all levels.
Communication of monitored results	The government stakeholders have been informed through the preparation and submission of a Monitoring Report.
Consultation records	Refer section 2.2 of the registered PD for details of consultation conducted prior to validation and records of local meetings conducted before distribution in monitoring period.
Stakeholder input	Refer section 2.2 of the registered PD for details of consultation conducted prior to validation.

2.1.3 Free, Prior, and Informed Consent

Consent	Prior to commencing the project, DelAgua signed a Memorandum of Understanding with Ministry of Environment to distribute improved cookstoves to rural households across Rwanda. Following this, DelAgua has signed Collaboration Agreements with the districts that the project has been delivered in (to date.) These agreements help the population in protecting the environment by scaling down firewood consumption while reducing indoor air pollution leading to improved respiratory health. Prior to distribution all potential ICS recipients are gathered to introduce the project. They are invited to receive a ICS free of
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	charge. All recipients must consent to receive an ICS by attending a distribution day and signing the agreement on the bottom of the DelAgua poster.
Outcome of FPIC	Not applicable, the project activity involves distribution of improved cookstoves to individual households and hence any possibility of forced eviction/physical relocation of people due to the project is not applicable. There are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), or customary rights holders. It is a completely voluntary activity and households participating are free to choose whether they take part or not. The PP collaborates with the local entrepreneurs for fabrication as well as installation of project ICS thereby empowering them economically rather than relocation / threat.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
No major grievances have been received during the monitoring period.	Not applicable
The PP has established a comprehensive grievance redressal mechanism to ensure that stakeholders have access to a transparent and easily accessible system for ongoing consultation and resolution of project-related concerns. Key aspects of the process are as follows: A. Grievance Registration: Stakeholders, including end users, can raise grievances through multiple channels: - Submitting grievances via the company's website: https://www.delagua.org/Contactus/ - Raising issues in person at the field level, which are then escalated by the relevant field personnel to the designated grievance manager. - Calling the toll-free helpline dedicated to grievance reporting. B. Field-Level Escalation Mechanism: During the onsite visit, stakeholders confirmed that Community Health Workers (CHWs), who serve as field staff and local resource persons, act as intermediaries for reporting grievances. These individuals escalate concerns raised by project beneficiaries within their zones. As confirmed by households during the visit, DelAgua conducts regular household visits—aiming for at least one per year—during which staff engage directly with stakeholders and ensure grievances are documented and addressed. C. If a grievance is brought to a local staff member, they are instructed to escalate it to their line manager, who logs the grievance at the local office. D. Grievance Logbook and Periodic Visits:	

- A grievance logbook is maintained at the local office in Kigali, Rwanda.
- Field staff conduct regular visits to project households, providing an additional opportunity for stakeholders to communicate concerns directly.

These procedures support effective communication, ensure continuous engagement with stakeholders, and reinforce the project's commitment to transparency, inclusivity, and timely resolution of concerns.

Please see submitted grievance register for record of minor grievances raised during this MP.

2.1.5 Public Comments

Summary of comments received	Actions taken
No comments were received.	Not applicable

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

DelAgua has been delivering cookstove projects for over 10-years and has several active projects listed under VERRA. DelAgua does not sub-contract any of the implementation of this project to non-DelAgua entities. Thus, DelAgua management teams have expertise or experience in implementing similar project activities and engaging communities.

The operational management team for this project are Rwandan nationals with deep experience of rural delivery of services and government engagement.

Senior management of DelAgua manage multiple projects across multiple jurisdictions with deep experience in Carbon Methodologies, Research, Data Analysis and management of large scale development projects. DelAgua staff are well-trained, with prior experience in distribution and monitoring. They received training on the questionnaire before conducting the surveys.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	Natural Threat: The climate crisis and natural disasters can precipitate displacement and migration, resulting in the relocation of households.	The grouped project involves distribution of ICS that are portable and lightweight. Consequently, users of the project's ICS can easily carry them in case relocation becomes necessary.
Risks to stakeholder participation	N/A	N/A

Working conditions	DelAgua staff at risk from working in remote locations	DelAgua maintains a health, safety & security policy for all staff in Rwanda. In addition, DelAgua maintains a risk register (as part of an overall risk management policy) that tracks all risk types (in addition to safety and security.) This is updated on a quarterly basis.
Safety of women and girls	Women and girls are exploited as a result of participating in the project	DelAgua ensures all staff follow a clear Code of Conduct that they sign up to upon recruitment. This policy is regularly updated and enforced. Toll Free number exists within Rwanda and a clear whistleblowing policy is adhered to for any grievances raised. Training includes modules on safety and security. All surveys must first gain consent of the respondent. This individual must be above 18 or the survey is immediately ended.
Safety of minority and marginalized groups, including children	Children are recruited to deliver the project.	DelAgua maintains a policy on child labour which all staff, suppliers and vendors are required to adhere to. This policy is regularly updated and enforced.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risk identified	N/A, the project does not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous materials.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified⁶	Mitigation or preventative measure(s) taken
Discrimination	Risk of discrimination based on gender, race, religion, sexual orientation or other habits	DelAgua maintains several HR policies for staff as well as a safeguarding policy for beneficiaries. These policies are maintained, monitored and

⁶ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		implemented by DelAgua's HR unit. No instances of discrimination were record during this monitoring period.
Sexual harassment	Risk of sexual harassment	DelAgua maintains several HR policies for staff as well as a safeguarding policy for beneficiaries. These policies are maintained, monitored and implemented by DelAgua's HR unit. No instances of sexual harassment were record during this monitoring period.
Gender equity in labor and work	Risk of gender inequalities in recruitment process	DelAgua maintains recruitment policies that specifically commit the organization towards gender equity. Where possible, DelAgua works to try to increase female recruitment into the organization
Forced labor	Risk of illegal act of recruiting, transporting, harboring, or receiving people through force, fraud.	DelAgua maintains policies on forced labour and requires that all staff, suppliers and vendors agree to adhere to these policies.
Child labor	Children are recruited to deliver the project.	DelAgua maintains policies on child labour and requires that all staff, suppliers and vendors agree to adhere to these policies.
Human trafficking	Risk of illegal act of recruiting, transporting, harboring, or receiving people through force, fraud.	DelAgua maintains policies on forced and child labour and requires that all staff, suppliers and vendors agree to adhere to these policies..

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project involves the distribution of improved cookstoves to individual households. In the context of this project, there are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights held by stakeholders, indigenous people (IPs), local communities (LCs), or customary rights

	holders. The primary stakeholders are given the opportunity to provide their details if they wish to receive a free improved cookstove. It is entirely their choice whether to sign up or to decline. Once they have signed up, stoves are distributed irrespective of tribal, ethnic, religious or political backgrounds of the beneficiaries. Beneficiaries are not forced or coerced to use the stoves during ongoing monitoring work.
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2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	As the project is focused exclusively on the free provision of improved cookstoves, there is no impact on cultural heritage as part of project activities. Primary stakeholders are invited to give their name during the beneficiary data collection progress if they wish to receive an improved cookstove. They are not compelled to do so, nor are they told to change their traditional practices beyond being encouraged to use their improved cookstove for future cooking events.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The GP and its PAIs involves distribution of improved cookstove to households and communities in Rwanda and it does not require acquisition of property, hence no impact on any property rights. Hence, this is not applicable.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	Not applicable. The project activity involves distribution of improved cookstoves to individual households and it does not require acquisition of property.
Benefit sharing during the monitoring period	Not applicable.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified.	The ICS reduces GHG emissions through its better design and improved performance as compared to traditional stove, without changing the baseline fuel or

		introducing additional pollutants. Therefore, the project has no impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified.	The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. Thus, project activity will indirectly avoid erosion associated with tree cutting/ felling.
Water consumption and stress	No risk identified.	The by-products (ash and smoke) of the project's ICS match the baseline scenario, as there's no fuel change. This prevents any additional impact on natural watercourses, groundwater, or watersheds, including seasonal flow variations, flooding risks, aquatic connectivity issues, or water scarcity.

2.4.1 Rare, Threatened, and Endangered species

Not applicable, the project is not located in, or adjacent to habitats for rare, threatened, or endangered species.

2.4.2 Introduction of species

Not applicable, the project does not involve introduction of any species.

2.4.3 Ecosystem conversion

Not applicable, the project is not an ARR, ALM, WRC or ACoGS project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The first ICS under the grouped project has started distribution from 04 November 2022. The implementation status of the first project activity (PAI1) at the end of the monitoring period is as follows:

ICS Type	2022	2023	Total (Number)
BURN Kuniokoa Generation 2 (with pot-skirt)	132,228	202,716	334,944
Total	132,228	202,716	334,944

The stoves are deemed under continuous operation since their installation. In total 334,944 ICS have been installed since the beginning of the project. No events occurred during the monitoring period that could impact the GHG emission reductions.

3.2 Deviations

3.2.1 Methodology Deviations

There are no methodology deviations in this monitoring period.

3.2.2 Project Description Deviations

During the current monitoring period, the project has undergone a methodology change following the "Procedure to Change Methodology through a Project Description Deviation v4.0." The previously applied methodology, VMR0006 ver 1.1, has been replaced by VM0050 ver 1.0 for this monitoring period.

Impact on Methodology Applicability, Additionality, and Baseline Scenario

The project deviation does not impact the additionality of the project. The project continues to adhere to the same additionality approach by utilizing the positive list provided in the methodology, thereby maintaining its eligibility as additional.

All applicability conditions of the newly adopted methodology, VM0050 ver 1.0, have been duly justified in Section 3.2 of the revised Project Description (PD) and further elaborated in Section 3.4 of this report. The PD has also been updated to reflect relevant changes in methodology in Sections 4.0 and 5.0, aligning them with VM0050 ver 1.0 requirements. Correspondingly, this monitoring report has been updated to ensure consistency with the revised PD and the applied methodology.

The baseline scenario has been reassessed in accordance with the new methodology. However, the fundamental baseline scenario remains unchanged, which is the continued use of traditional cookstoves, such as three-stone fire cookstoves, relying on non-renewable wood fuel by the target population in rural and semi-urban areas. These traditional cookstoves fulfill similar thermal energy needs as the project cookstoves in the absence of the project activity.

Outcome of the Deviation

As a result of the reassessment, the baseline Kitchen Performance Test (KPT) results indicate a reduction in baseline fuel consumption from the previously determined 4.41 tons/year to 4.04 tons/year. This reduction represents a conservative approach, ensuring that the baseline emissions are not overestimated. The updated value reflects the revised methodology's approach to baseline determination while maintaining alignment with the project's overall objectives and operational framework.

3.3 Grouped Projects

No new project activity instances have been added in this monitoring period.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☒ Yes

☐ No

Use and Applicability of the Latest Approved Version of the Methodology

The project has undergone a methodology change in accordance with the "Procedure to Change Methodology through a Project Description Deviation v4.0." The previously applied methodology, VMR0006 ver 1.1, has been replaced with the following:

Reference ID, if applicable	Title	Version
VM0050	VM0050 - Energy efficiency and fuel-switch measures in cookstoves	1.0

All applicability conditions of the newly adopted methodology, VM0050 ver 1.0, have been duly justified in Section 3.2 of the revised PD.

Updates to the Project Description Reflecting the New Baseline

The baseline scenario has been reassessed in accordance with the new methodology. However, the fundamental baseline scenario remains unchanged. The continued use of traditional cookstoves, such as three-stone fire cookstoves relying on non-renewable wood fuel, remains the primary cooking method for the target population in rural and semi-urban areas. These traditional cookstoves fulfill similar thermal energy needs as the project cookstoves in the absence of the project activity.

This assessment has been supported by a KPT conducted in baseline households. According to the baseline KPT, all 128 surveyed households used a three-stone fire as their primary cooking stove. The results were further corroborated by literature and reports cited in Section 3.4 of the PD. The KPT was conducted in accordance with the protocol and the requirements of the applied methodology, as detailed in Section 3.4 of the PD.

Validity of the Baseline Scenario

The reassessment confirms that the baseline scenario remains valid under the revised methodology. The baseline KPT results indicate a reduction in baseline fuel consumption from the previously determined 4.41 tons per household per year to 4.04 tons per household per year. This updated value aligns with the revised methodology's approach to baseline determination while maintaining consistency with the project's overall objectives and operational framework. The percentage change between the previously determined and revised baseline fuel consumption values is approximately 8%.

Impact of National and Sectoral Policies on Baseline Validity

No significant national or sectoral policy changes have been identified that would impact the validity of the baseline scenario. The reassessment, conducted in accordance with the requirements of the newly adopted methodology, confirms that the baseline scenario remains relevant and applicable. The continued reliance on traditional biomass cookstoves in the target population, as confirmed through field studies and literature, supports the applicability of the reassessed baseline scenario under the new methodology.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{b,i,CO2}$ $EF_{p,j,CO2}$
Data unit	t CO ₂ /TJ
Description	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario CO ₂ emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 112 t CO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 112 t CO ₂ /TJ
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	$EF_{b,i,non-CO2}$ $EF_{p,j,non-CO2}$
Data unit	t CO ₂ e/TJ
Description	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario Non-CO ₂ emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 9.46 t CO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 9.46 t CO ₂ e/TJ (AR5 GWP)
Purpose of data	Calculation of baseline and project emissions

Comments	N/A
Data / Parameter	NCV _{b,i} NCV _{p,j}
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type <i>i</i> Net calorific value of project fuel used by project device type <i>j</i>
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 0.0156 TJ/tonne
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 0.0156 TJ/tonne
Purpose of data	Calculation of baseline and project emissions
Comments	N/A
Data / Parameter	BC _{ex-ante,b,i} ⁷
Data unit	Tonnes/year
Description	Ex-ante annual average quantity of fuel used per baseline device type <i>i</i>
Source of data	Option 1: A measurement campaign (Baseline KPTs), conducted following the Kitchen Performance Test Protocol protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Value applied	4.04
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante. A measurement campaign (Baseline KPTs), conducted following the Kitchen Performance Test Protocol protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.

⁷ This parameter corresponds to BC_{ex-ante,b,i} for the first two years and where the follow-up baseline survey campaign shows that there are no significant changes in baseline fuel consumption. Otherwise, it corresponds to BC_{b,i,y}.

	The survey revealed that the average wood consumption in the Rwanda is 0.937 tonnes/person/year. This result is scaled appropriately using the average household size (H_{hi}) as follows: $BC_{ex-ante,b,i} = BC_{ex-ante,b,l} \text{ (tonnes/person/year)} \times \text{Equivalent standard male adults } (H_{hi}) \text{ obtained from baseline KPT}$ $= 0.937 \times 4.31$ $= 4.04 \text{ tonnes/year}$ The details of the baseline KPTs including target population, sample size, sampling method, precision/confidence achieved has been added in section 3.4 of the PD.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction
Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	VM0050 Default value
Value applied	0.15
Justification of choice of data or description of measurement methods and procedures applied	This data is fixed ex-ante as per the applied methodology VM0050. A default value of 0.15 shall be used if the baseline device is a three-stone fire using firewood or a cookstove without improved combustion air supply or flue gas ventilation. The project distributes non-renewable wood fuel-based ICS, replacing only baseline devices that meet these criteria.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	H_{hi} $H_{hj,k}$
Data unit	Equivalent standard male adults
Description	Average household size of the target population using device type i Average household size of the target population using device type j from batch k

Source of data	Baseline survey /KPT
Value applied	4.31
Justification of choice of data or description of measurement methods and procedures applied	A measurement campaign (Baseline KPTs/Surveys), conducted following the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Purpose of data	Estimation of average energy consumption when applying Option 1: Measurement campaign. Cross-checking energy and fuel consumption values.
Comments	N/A

4.2 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	The following data is recorded during project activity implementation: 1) Number of new devices distributed under the project activity, identified by the type of device and date of commissioning; and 2) Identification information of the recipient of the device distributed under the project activity (e.g., name, address, phone number). Data management and reporting of this information must adhere to both data privacy requirements and good practice.
Frequency of monitoring/recording	Every time that new project devices are distributed
Value applied	334,944
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	The number of project devices will be recorded in a database, sales record, or similar to ensure transparency. Here, the batch k

	is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.
Data / Parameter	$n_{j,k,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Option 2 (surveys): Based on an adoption rate determined by a survey according to the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities.
Frequency of monitoring/recording	Option 2 (surveys): annually
Value applied	0.9 Although the operational check was 99.08% as per the survey, in line with the clarification on VM0050, a cap of 0.9 was applied. 1) The PP has capped the usage rate at 90% in line with the VM0050 v1.0 Corrections and Clarifications Number 2, based on the following: <i>Selection of technologies and fuels that fully meet the cooking needs of the target population demonstrated by citing robust research or conducting an investigation of cooking practices and attitudes</i> - As detailed in Section 3.4 of the PD, the target population—rural households in Rwanda—primarily use wood fuel and traditional wood stove technology for cooking. This is supported by both the project's baseline survey and national statistics (the Fifth Rwanda Population and Housing Census, 2022). - The PP has distributed the Kuniokoa stove, an improved wood-burning stove, specifically to replace the traditional 3-stone fires or traditional wood stoves used by these households. This ensures that the distributed technology and fuel type fully meet the cooking needs of the target population. <i>Implementation of support activities to assist households in effectively operating and maintaining their cookstoves. and maintaining their cookstoves. These may include providing materials (print, in-person, or video) on how to operate the cookstove to prepare common local foods, how to troubleshoot common operational issues, and how</i>

	to make minor repairs (including obtaining necessary replacement parts). All such communications and materials must be provided in local language(s) commonly used in the project area. ○ At the time of distribution, each household receives a poster agreement with pictorial representations showing how to properly use and maintain the stove, including clear do's and don'ts. These instructions are provided in the local language, ensuring accessibility and understanding. Demonstrations are also conducted during distribution events. ○ Additionally, during follow-up visits, local field teams provide users with training using a 16-page Tubebo Neza Flipbook, which covers proper usage, maintenance, and troubleshooting. This material is also in the local language. The flipbook and related training content were reviewed and verified by the VVB and are included as Appendix 2 of FVR Document No. 29. 3. <i>Provision of a commonly used, toll-free communications channel through which the target population can contact the project proponent to access support (e.g., maintenance and repair services)</i> ○ As detailed in Sections 2.1.4 of the PD and MR, the PP has established a toll-free helpline number, enabling end users to easily reach out with feedback, requests for maintenance, or any grievances. Considering all these points, the project fully complies with all three conditions set out under Clarification Number 2 of VM0050 v1.0 Corrections and Clarifications, thereby justifying the capping of the usage rate at 90%.
Monitoring equipment	N/A
QA/QC procedures to be applied	The date on which a sample project device stopped being used should be taken as follows: Option 2 (surveys): Where the project device is not working or not being used at the time of conducting the survey, it should be conservatively assumed that the project device has not been active since the date on which the last adoption survey was conducted.
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	The batch k is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction or %
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y
Source of data	Third party report in line with TOOL30 version 4.0 and applying an uncertainty discount of 26%
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Ex-ante for each crediting period
Value applied	0.6565
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	The value of 0.8872 was registered in the PD under TOOL30 Version 4 for the host country, Rwanda. This registered value in the PD is then adjusted by an uncertainty discount of 26% as follows: $= 0.8872 \times (1-26\%) = 0.6565$
Comments	This data has been fixed for entire crediting period.

Data / Parameter	$BC_{b,i,y}$
Data unit	Tonnes/year
Description	Fuel used per baseline device type i during year y
Source of data	Option 1: A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption
Description of measurement methods and procedures to be applied	Option 1: Follow-up baseline surveys must be conducted at most every five years from the date of the last survey in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption.

	Measurement campaign must be updated where follow-up baseline surveys show that the fuels, fuel sources, or technologies used by the control group are no longer statistically equivalent to the pre-project conditions of project households. The measurement campaign must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The result must be scaled appropriately using the average household size to obtain the value of $BC_{b,i,y}$.
Frequency of monitoring/recording	At most every five years
Value applied	4.04
Monitoring equipment	N/A
QA/QC procedures to be applied	The campaign achieved a confidence and precision of 95/10 for the target parameter of average daily fuel consumption per adult equivalent.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	The efficiency must be established using one of the following methods, and the corresponding documentation must be presented: 1) Water Boiling Test campaigns achieving 90/10 confidence and precision levels as per the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities; 2) Manufacturer-certified value that is determined via Water Boiling Test; or 3) Certification from the host country's national standards body or certifying agency.

	The decrease in thermal efficiency of project device j from batch k due to aging is accounted by conducting Standard Water Boiling Test campaigns during the monitoring period.	
Frequency of monitoring/recording	Annually	
Value applied	Kuniokoa Efficiency value	
	With pot skirt	Without pot skirt
	0.4861	0.4095
	Monitored Utilization of Pot-skirt	Final Weighted Average $\eta_{new,i,j}$
	89.57%	0.4781
Monitoring equipment	N/A	
QA/QC procedures to be applied	N/A	
Purpose of data	Calculation of baseline and project emissions	
Calculation method	N/A	
Comments	-	

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes/year
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Option 1: Kitchen Performance Test (KPT) A measurement campaign following the Kitchen Performance Test Protocol protocol was designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The campaign achieved confidence and precision of at 95/10 for the target parameter of average daily fuel consumption per adult equivalent. The result was scaled appropriately using the average household size to obtain the value of $BC_{p,j,k,y}$. Project KPTs also measured the quantity of fuel used by pre-project devices that are still operational and provide it in the calculation sheet and monitoring reports (this is a reporting-only requirement).
Frequency of monitoring/recording	Biennial or annual for Option 1
Value applied	1.21

Monitoring equipment	N/A
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline and project emissions
Calculation method	A measurement campaign (Project KPTs), conducted following the Kitchen Performance Test Protocol protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The survey revealed that the average wood consumption on the project ICS is 0.282 tonnes/person/year. This result is scaled appropriately using the average household size ($Hh_{j,k}$) as follows: $BC_{p,j,k,y} = BC_{p,j,k,y} \text{ (tonnes/person/year)} \times \text{Equivalent standard male adults } (Hh_{j,k}) \text{ obtained from project KPT}$ $= 0.282 \times 4.31$ $= 1.21 \text{ tonnes/year}$
Comments	N/A

4.3 Monitoring Plan

This Grouped Project in *Rwanda* follows VCS standard requirements applicable, along with the methodological requirements on monitoring a grouped non-AFOLU large project activity. The Monitoring Plan is elaborated in multiple sub-sections for the sake of clarity. This monitoring plan is based on the applied methodology VM0050 ver 1.0. The monitoring plan consists of checking a representative sample of all ICS units distributed to ensure that they are still operating or are replaced by an equivalent in-service unit.

1. Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters

A project database was populated during the distribution of each initial cookstove issued. The database is accessible to the project proponent, appropriate partners, and the verification VVB. The database includes, at minimum, the following:

- Date of Distribution
- Geographic area of distribution;
- Model/type of project technology distributed
- Name, telephone number (where available), and address of recipient; and
- Unique alpha/numeric ID for each device that is distributed

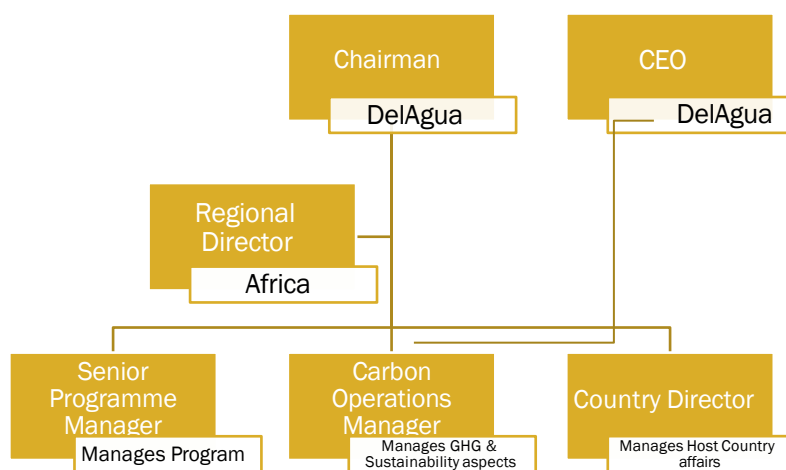
The database is used to select a random, representative sample for monitoring and verification purposes. The data was collected using surveys carried out by project proponent own teams. Monitoring surveys includes examination of the cookstove installations and verification that they are in an operational condition. Data collected from the surveys was compiled into an Excel spreadsheet. Data obtained from the samples were used to estimate proportions and mean values for the parameters described in below sub-sections. The values then factored into the emissions reduction calculations.

The data recorded in the monitoring surveys is collected and stored electronically, and then uploaded to the central database management system, where it is maintained online on a subscription-based server host.

The data will be also stored offline, within the hard disk drives. The data collection procedures do not involve any paper records, to reduce the environmental footprint of the programme.

The project related data will be stored for two years after the implementation period is over for the project activity.

2. Organizational Structure, Role, and Responsibilities of Personnel Involved in the Project Activity



The DelAgua team involved in the GHG operations management is the local staff based in Rwanda along with the senior management.

The key responsibilities are shared by the team as follows:

Chairman – Supervises the overall programme at strategy and financial side.

COO – Acts as central point of strategy, programme management, finance, and technical domains

Director Business Intelligence – Manages the programme at implementation level, and acts as an internal technical reviewer for the programme documents and procedures.

Carbon Operations Manager – Manages the operations for GHG emission mitigation programme, from planning to implementation stage. Additionally, manages the Sustainability aspects of the programme.

Country Director – Supports programme in the operations and management at the host country level and acting as point of communication between DelAgua and government of *Rwanda*.

The collective responsibilities of DelAgua as the Project Participant is following:

- Acts as the project managing entity
- Development of PD
- Registration/Issuances of the project with VERRA
- Inclusion of new PAIs upon satisfaction of the eligibility criteria stipulated in the PD
- Authorized entity for any official communication with the VERRA and VVB
- Manage and archive the database for all appliances
- QA/QC of database during distribution & data collection
- Maintain local repair/replacement facilities
- Education and public awareness raising during initial distribution as well as periodic education campaigns.
- Data collection during distribution & delivery of data to database
- QA/QC of data collection
- Conduct ongoing monitoring, as required to ascertain monitored parameters.

3. The procedures for Internal Auditing and QA/QC

Quality assurance and quality control (QA/QC) measures were adhered as described in section 5.2 below. Distributors are trained in the proper installation, use, and maintenance of the devices, and are provided with training sufficient to introduce recipient households to the proper operation and maintenance of the devices, as well as contact information for DelAgua. Supervisors receive similar training, as well as supervisory training in monitoring of distributor's activities. Supervisors are responsible for approximately 10 distributors. During the campaign, DelAgua staff spot-checks distributor quality, by revisiting 2 households per distributor, to verify that the stoves are in place, and that the recipient can show proper operation of the units. The results of this spot check are recorded electronically and compared against the initial distribution logs. The data collected in the database is verified and controlled in the following way:

- Surveys are conducted by staff trained by DelAgua.

- Supervisors are separately trained by DelAgua to spot-check surveyor results.
- All ex-ante and monitored data parameters are collected on smart phone-based surveys and are uniform across all surveyors for any given survey activity.
- Unknown to the surveyors, the database analysis includes verification of “valid” surveys including by a.) verifying household consent is received and b.) that the survey duration, as recorded by Internet time, is no shorter or longer than a reasonable value.
- Data validation is conducted on the database to ensure that values are numerically consistent, i.e. multiple-choice values add up to 100% of respondents.

4. Procedures for handling non-conformances with Validated Monitoring Plan

The personnel engaged in monitoring and reporting underwent training to ensure a comprehensive understanding and proper execution of the monitoring and reporting procedures outlined in the monitoring plan. DelAgua is responsible for training any contractors used during distribution, education and monitoring activities. DelAgua ensures training of all on-site staff with respect to adherence to the Monitoring Plan of the project activity. Records of the training will be kept. Training records are saved for such training programs, where step-by-step training is provided to the surveyors for a quality data collection process, where they are expected to not impact the materially, degree of conservativeness, and perform the surveys in an unbiased manner. The surveyors are trained on a periodic basis, and refresher trainings are also provided to improve the quality of surveys from time to time. During these audits all the data and parameters that need to be monitored as per the monitoring plan will be checked and shortcomings if any will be reported and addresses. Since the surveys are electronic in format, recording of responses and the record keeping procedure for these surveys are conveyed to the team through rigorous trainings.

The project proponent ensured that field personnel have reviewed, understood and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. Monitoring plan explains how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household is chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires is used.

5. Sampling Plan:

Due to the large number of ICS distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling have been undertaken as part of Sampling Plan in line with the requirements of the methodology applied and the Standard: Sampling and surveys for CDM project activities and programmes of activities, version 9.0.

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.

STEP 1: Sampling Objectives

The sampling objective for each parameter was to determine, via sampling survey, a statistically significant parameter value for the emission reduction calculations. These parameters are as listed as below:

No.	Parameter	Description	Parameter type	Monitoring frequency
1	$\eta_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that remain operating in year y	Proportion	Annually
2	$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y	Mean	Annually
3	$BC_{b,i,y}$	Fuel used per baseline device type i during year y	Mean	Every five years
4	$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y	Mean	Biennial or annual

STEP 2: Target Population

Total 334,944 households are part of the PAI-1 till the end of current monitoring period, were considered for sampling.

STEP 3: Sampling Frame

The sampling frame refers to all the aggregated data of end-users of the ICS as recorded in the Project ICS Database. The monitoring survey data was collected through physical survey which is the primary mechanism in place for the data collection of the newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) which is used in the current monitoring period.

The following information was recorded during the monitoring survey, at a minimum:

- Product based Unique Identification number.
- Contact information, geographical location or other relevant information of the end user
- Sampling frame was representative of the entire population.
- No elements from outside the population of interest is present for the current monitoring period. Wherever information miss-match is observed, it is corrected in the database and conservative assumptions is applied.

STEP 4: Sample Method

Sampling was conducted using stratified random sampling technique over the sampling frame, and detailed calculations are provided below as per CDM guidelines “Sampling and surveys for CDM project activities and programmes of activities” v 9.0. The sampling was done on 10th December 2024.

To undertake random sample selection in a manner that accommodates the logistical challenge posed by the potential inclusion of ~ 100 (numbers corresponds to sample size) distinct villages and sectors, a simple random sampling method is deemed impractical. Therefore, a stratified random selection approach is employed. Initially, villages are randomly selected with the number required achieved by dividing the sample size by 5, where 5 corresponds to the anticipated number of monitoring surveys achievable in a day by a DelAgua staff member. This choice is based on the operational feasibility of one staff member being able to conduct surveys in one village per day.

This is randomly selected through the excel formula as follows:

`=INDEX (SORTBY (data,RANDARRAY(ROWS(data))),SEQUENCE(#villagesrequired))`

Where first dataset is the list of villages. This gives a list of random villages.

After the identification of random villages, all stoves distributed within the villages are compiled into a singular document. Each stove is assigned a unique identifier, following which, using the =randbetween function, five stoves (households) are randomly selected from each of the chosen villages. This process gives the final sample set.

To determine the parameters, sampling involves the following approaches.

$\eta_{j,k,y}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$\eta_{new,avg,y}$: WBT tests were conducted in the sampled households.

$BC_{b,i,y}$: Baseline KPTs were conducted in the sampled households

$BC_{p,j,k,y}$: Project KPTs were conducted in the sampled households

STEP 5: Desired Precision / Expected Variance and Sample Size

To estimate the number of premises to be surveyed with a 10% margin of error at desired confidence level of 95%, the optimal sample size n is given by:

$$n \geq \frac{z^2 * N * V}{(N-1) * precision^2 + z^2 * V}$$

Where:

Parameter	Description
-----------	-------------

n	Sample Size
N	Total number of cookstoves (ICS) distributed
Z	Constant referring to level of confidence (1.96 for 95 %, 1.646 for 90%)
Precision	Required precision (e.g., 10% = 0.1)

Where for Proportion Parameter:

$$V = \frac{SD^2}{p^2}$$

$$SD^2 = \frac{\sum_{i=1}^k g_i * p_i * (1 - p_i)}{N}$$

$$\bar{p} = \frac{\sum_{i=1}^k g_i * p_i}{N}$$

Parameter	Description
SD	Standard deviation
g _i	Weight of strata i in the population
p _i	Expected proportion of strata i in the population
k	Total number of strata in the population

And for mean-based parameters:

$$V = \frac{SD^2}{mean^2}$$

$$SD^2 = \frac{\sum g_i \times SD_i^2}{N}$$

$$mean = \frac{\sum g_i \times m_i}{N}$$

Parameter	Description
Mean	Weighted overall mean
m _i	Mean of strata i

As per the CDM guideline for Sampling and survey⁸, if the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen. The sample size calculation and monitoring results of the monitoring parameters for PAI-1 has been shown below:

⁸ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

Sampling Constants	Values
Monitoring period start	1/1/2023
Monitoring period end	31/12/2024
Level of Sampling	Project Level
Monitoring Frequency	Annual
Confidence (%) (90 or 95)	95%
Margin of Error (%)	10%
Z value	1.960

- Parameter ($\eta_{new,avg,y}$)**

As per the sample size calculation, 8 samples were calculated to be surveyed, However, 13 samples were considered to compensate for any outliers or non-responses associated with samples and to get accurate results. The snapshots below demonstrate that required precision is met i.e., the parameter is within the 95/10 confidence/precision. For detailed calculation of the standard error and relative precision of the parameter $N_{y,i,j}$, refer tab “Sample Size Cal & Results” of the ER sheet.

Monitoring parameter(s)		Project Stove (Kuniokoa) Efficiency $\eta_{new,avg,y}$							
Sampling frame(s)		Given stove model and vintage population							
Sampling approach		Stratified random sampling							
Sampling frame	Sampling Population	Expected Value (without pot-skirt)	Expected SD (without pot-skirt)	Expected Value (with pot-skirt)	Expected SD (with pot-skirt)	Calculated Minimum Sample Size without pot-skirt (n)	Calculated Minimum Sample Size with pot-skirt (n)		
2022	132228	0.43	0.043	0.51	0.051	3	3		
2023	202716	0.43	0.043	0.51	0.051	5	5		
		Estimated efficiency (mean)				0.43	0.51		
		Estimated Standard Deviation of efficiency (SD)				0.04	0.05		
		$V_{mean} = (SD/mean)^2$				0.01	0.01		
		Minimum Sample Size required (efficiency)				4	4		
tDistribution sample size adjustment		Iteration 1				11	11		
		Iteration 2				5	5		
		Iteration 3				8	8		
		Iteration 4				6	6		
		Iteration 5				7	7		
		Iteration 6				6	6		
		Iteration 7				7	7		
Monitoring results									
Sampling Population	Sampling frame size	Monitored Sample Size (without pot-skirt)	Monitored Efficiency (Fraction) (without pot-skirt)	Monitored Standard Deviation (without pot-skirt)	Monitored Sample Size (with pot-skirt)	Monitored Efficiency (Fraction) (with pot-skirt)	Monitored Standard Deviation (with pot-skirt)		
2022	132228	5	0.4055	0.00589	5	0.4825	0.00969		
2023	202716	8	0.4121	0.00666	8	0.4885	0.00760		
Reliability Check									
Samples Monitored				13	Samples Monitored				13
Efficiency Monitored				40.95%	Efficiency Monitored				48.61%
Standard error of mean				0.0018	Standard error of mean				0.0024
Relative precision (Margin of error) (%)				0.938%	Relative precision (Margin of error) (%)				1.058%
Result				Ok, reliability level met	Result				Ok, reliability level met
Lower Bound confidence value				Not applicable	Lower Bound confidence value				Not applicable

- Parameter ($\eta_{j,k,y}$)**

Monitoring parameter(s)		Stove Operating Fraction for determination of $n_{j,k,y}$	
Sampling frame(s)		Given stove model and vintage population	
Sampling approach		Stratified random sampling	
Sampling Population	Sampling frame size	Expected operational proportion (SoF)	Calculated Minimum Sample Size (n)
2022	132228	0.90	17
2023	202716	0.90	26
Sample size determination			
Estimated SOF (p)			0.90
Estimated Standard Deviation of SOF (SD)			0.30
$V_{SOF} = (SD/p)^2$			0.11
Sample Size required (SOF)			43
Monitoring results			
Sampling Population	Sampling frame size	Monitored Sample Size ($n_{j,ktotal}$)	Monitored Operating Fraction ($n_{j,koperational}/n_{j,ktotal}$)
2022	132228	60	1.00
2023	202716	66	0.98
Reliability Check			
Samples Monitored			126
SoF Measured			99.08%
Standard Error of SoF			0.910%
Relative precision (Margin of error)			1.80%
Result			Ok, reliability level met
Lower Bound confidence value			Not applicable

As per the sample size calculation, 43 samples were calculated to be surveyed, However, 126 samples were considered to compensate for any outliers or non-responses associated with samples and to get accurate results. The snapshots above demonstrate that required precision is met i.e., the parameter is within the 95/10 confidence/precision. For detailed calculation of the standard error and relative precision of the parameter $N_{y,i,j}$, refer tab “Sample Size Cal & Results” of the ER sheet.

- Parameter ($BC_{b,i,y}$)

Refer section 3.4 of the PD for the sample size of this mean parameter and results.

- Parameter ($BC_{p,j,k,y}$)

As per the sample size calculation, 47 samples were calculated for this mean parameter as per CDM sampling standard, However, 116 samples were considered for surveys. There were two reasons- firstly, common survey was conducted for $n_{j,k,y}$ and $BC_{p,j,k,y}$ and secondly, to compensate for any outliers or non-responses associated with samples. The snapshots above demonstrate that required precision is met i.e., the parameter was within the 95/10 confidence/precision. For detailed calculation of the standard error and relative precision of the parameter $BC_{p,j,k,y}$, refer project KPT analysis sheet.

Kg/person/day assessment	Project KPT
Number of Samples	116
Median	0.76
Quartile (Q1)	0.57
Quartile (Q3)	1.01
IQR	0.44
Upper Quartile limit	1.66
Lower Quartile limit	0.00
95-10 Precision Assessment	Project KPT
Average consumption (kg/person/day)	0.77
Standard Deviation	0.31
Coefficient Of variance (CV)	40.79%
Lower Bound value of confidence interval	0.71
Upper Bound value of confidence interval	0.83
Confidence interval	0.11
Precision (2-sided)	7.42%
95-10 Rule Met?	YES
Project Stove Consumption BCp,j,k,y (tons/person/yr)	0.282
Project Stove Consumption BCp,j,k,y (tons/HH/yr)	1.213

STEP 6: Quality Assurance/Quality Control

The project proponent applied a confidence/precision of 95/10 for each monitored parameter allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to both parameters i.e. number of operational cookstoves, and adjustment factors. The project proponent monitored a particular parameter till the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the applied methodology VM0050, ver. 1.0, baseline emissions are calculated as follows:

$$BE_y = \sum EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2}) \quad \text{Equation (1)}$$

Where:

BE_y	Baseline emissions during year y (t CO ₂ e)
$EC_{i,y}$	Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	Number of commissioned project devices of type j from batch k in year y

$\eta_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
EF_{b,i,CO_2}	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ /TJ)
$f_{NRB,y}$	Fraction of woody biomass that is established to be non-renewable used by baseline device in year y (fraction)
$EF_{b,i,nonCO_2}$	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ e/TJ)
i	Baseline device type and its respective fuel type
j	Project device type and its respective fuel type

The average energy consumption of baseline device type i is calculated as follows:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad \text{Equation (2)}$$

Where:

$BC_{b,i,y}$	Fuel used per baseline device type i during year y (tonnes)
$NCV_{b,i}$	Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

Cross-check of $EC_{i,y}$ to address stove stacking:

Baseline consumption from the back calculation is calculated as follows:

$$EC_{est,y} = EC_{p,y} \times \eta_{new,avg,y} / \eta_{old,avg} \quad \text{Equation (3)}$$

Where:

$EC_{est,y}$	Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year y (TJ)
$EC_{p,y}$	Energy used in project scenario by project devices during year y (TJ)
$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y (fraction)
$\eta_{old,avg}$	Weighted average efficiency of baseline devices that are replaced by project devices (fraction)

Note: Where the results indicate that baseline consumption ($EC_{i,y}$) is higher than that indicated by back-calculation from the project scenario ($EC_{est,y}$) then the back-calculation results ($EC_{est,y}$) will be applied in Equation (1) as a conservative cap.

$$EC_{p,y} = \sum_{j,k} BC_{p,j,k,y} \times NCV_{p,j} \quad \text{Equation (4)}$$

Where:

$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y (tonnes or m ³)
$NCV_{p,i}$	Net calorific value of project fuel used in project device type i (TJ/tonne or TJ/m ³)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, Equation (5) of the methodology applies to electric stoves and is therefore not applicable.

5.2 Project Emissions

As per the applied methodology VM0050, ver. 1.0, project emissions are calculated as follows:

$$PE_y = PE_{energy,y} + PE_{others,y} \quad \text{Equation (6)}$$

Where:

PE_y	Project emissions during year y (t CO ₂ e)
$PE_{energy,y}$	Project emissions from energy consumption of project devices in year y (t CO ₂ e)
$PE_{others,y}$	Project emissions from other sources in year y (t CO ₂ e)

$$PE_{energy,y} = \sum_j \sum_k BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times (EF_{p,j,CO2} \times f_{NRB,y} + EF_{p,j,nonCO2}) \quad \text{Equation (7)}$$

Where:

$EF_{p,j,CO2}$	CO ₂ emission factor for fuel used by project device type j in the project scenario (t CO ₂ /TJ) = 112 tCO ₂ e/TJ (Meth Default Value)
$EF_{p,j,nonCO2}$	Non-CO ₂ emission factor for fuel used by project device type j in the project scenario (t CO ₂ e/TJ) = 9.46 tCO ₂ e/TJ (Meth Default Value)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, Equation (8) and Equation (10) of the methodology applies to electric stoves and are therefore not applicable. Further, the ICS distributed under the project is designed for wood fuel, and beneficiaries procure the wood fuel either from nearby fields or the market. Therefore, Equation (9) is not applicable.

5.3 Leakage Emissions

The value of Leakage has been considered as default of 0.95 in accordance with methodology.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

As per the applied methodology VM0050, ver. 1.0, Net GHG emission reductions are calculated as follows:

$$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y} \quad \text{Equation (11)}$$

Where:

ER_y Emission reductions during year y (t CO₂e)
 $LE_{RB,y}$ Leakage emissions associated with use of renewable biomass during year y (t CO₂e)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, leakage emissions associated with the use of renewable biomass are not applicable, as the project does not involve renewable biomass as a fuel source.

The calculation of ERs for MP-2 is as follows:

Parameter	2023	2024	Unit	Formula	Source
$f_{NRB,y}$	0.8872	0.8872	Fraction	-	Registered PD
$f_{NRB,adjusted}$	0.6565	0.6565	Fraction	-	Adjusted using default adjustment factor given in VM0050
$NCV_{b,i} = NCV_{p,i}$	0.0156	0.0156	TJ/tonne	-	VM0050 Default Value
$EF_{b,i,CO2} = EF_{p,i,CO2}$	112	112	tCO ₂ e/TJ	-	VM0050 Default Value
$EF_{b,i,non-CO2} = EF_{p,i,non-CO2}$	9.46	9.46	tCO ₂ e/TJ	-	VM0050 Default Value
LE	0.95	0.95	Fraction	-	VM0050 Default Value
$\eta_{old,avg}$	0.15	0.15	Fraction	-	VM0050 Default Value
$N_{j,k,y}$	334,944	334,944	Number	-	ICS Distribution Summary
$\eta_{j,k,y}$	0.9000	0.9000	Fraction	-	Ex-post Survey Results
% usage of pot-skirt	0.8957	0.8957	Fraction	-	Ex-post Survey Results
$\eta_{pskirt_Kuniokoa}$	0.4861	0.4861	Fraction	-	WBT Tests
$\eta_{pnon-skirt_Kuniokoa}$	0.4095	0.4095	Fraction	-	WBT Tests
$\eta_{new,avg,y}$	0.4781	0.4781	Fraction	-	Calculated
$Hh_i (Hh_{j,k})$	4.31	4.31	Number	-	Baseline KPT
$BC_{b,i,y}$	0.94	0.94	Tonnes/person/ year	-	Baseline KPT
$BC_{b,j,y}$	4.04	4.04	Tonnes/HH/year	-	Calculated
$BC_{p,i,k,y}$	0.282	0.282	Tonnes/person/ year	-	Project KPT
$BC_{p,j,k,y}$	1.21	1.21	Tonnes/HH/year	-	Calculated
$EC_{i,y}$	0.063	0.063	TJ	$EC_{i,y} = BC_{b,i,y} * NCV_{fuel}$	Calculated
$EC_{p,y}$	0.019	0.019	TJ	$EC_{p,y} = BC_{p,i,k,y} * NCV_{fuel}$	Calculated
$EC_{est,y}$	0.060	0.060	TJ	$EC_{est,y} = EC_{p,y} * \eta_{new,avg,y} / \eta_{old,avg}$	Calculated
-	0.94	1.00	Fraction	-	Calculated
BE_y	1,422,299	1,509,444	tCO ₂ e	$BE_y = \text{Min} (EC_{i,y}, EC_{est,y}) * N_{j,k,y} * \eta_{j,k,y} * (EF_{CO2} * f_{NRB,adjusted} + EF_{non-CO2})$	Calculated
PE_y	446,211	473,550	tCO ₂ e	$PE_y = EC_{p,y} * N_{j,k,y} * \eta_{j,k,y} * (EF_{CO2} * f_{NRB,adjusted} + EF_{non-CO2})$	Calculated
ER_y	927,283	984,099	tCO ₂ e	$ER_y = (BE_y - PE_y) * 0.95$	Calculated
Total Emission Reductions		1,911,382			

For detailed calculations, refer ER sheet.

Below table illustrates of emission reductions achieved over the monitoring period.

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Removal VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)

01-January-2023 – 31-December-2023	1,422,299	446,211	0	927,283	0	927,283
01-January-2024 – 31-December-2024	1,509,444	473,550	0	984,099	0	984,099
Total	2,931,743	919,761	0	1,911,382	0	1,911,382

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-January-2023 – 31-December-2023	1,045,259	927,283	~11% lower	The primary reason for lower VCUs in the current monitoring period attributed to lower ICS distribution than envisaged ex-ante and distribution of ICS in phases during current monitoring period. Hence, average operational age of ICS is significantly lower than number of days in current monitoring period, which is appropriate.
01-January-2024 – 31-December-2024	1,015,981	984,099	~3% lower	
Total	2,061,240	1,911,382	~7% lower	

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

No information presented in the monitoring report is deemed commercially sensitive.

APPENDIX 2: EVIDENCE OF PROJECT SD CONTRIBUTIONS

The SDG contributions for SDGs 1, 3, 5, 7, and 15 are calculated directly in the ER sheet based on survey data. Screenshot showing these calculations is attached here as evidence. For detailed information on how these contributions are calculated, please refer to the 'Ex-post Survey Results' tab in the ER sheet.

SDG Indicator	Monitoring Parameter		
1.4.1 Proportion of population living in households with access to basic services	Number of ICS distributed	334944	
3.9.1 Mortality rate attributed to household and ambient air pollution	The proportion of HHs confirming the reduction in smoke after using the project ICS (%)	99.21%	332,286
5.4 Proportion of time spent on unpaid domestic and care work, by sex, age and location	The proportion of HHs confirming the reduction in amount of time spent on collection of firewood and cooking time (%)	96.83%	324,311
7.1.2 Proportion of population with primary reliance on clean fuels and technology	The Number of HHs with operational ICS	331,873	
15.4.2 Mountain Green Cover Index	Amount of firewood saved by the project (tons)	1,554,060	1,980,628

Evidence for SDG 8 (employment list) and SDG 4 (training records) is shared directly with the VVB, which involves numerous pages. Including them here would add at least 50-60 pages to the MR, so to keep the document manageable, this information is shared separately.